AMD Ryzen 5000 Performance
Leaks suggest AMD Ryzen 5000 based laptops out-perform Intel's offering in multi-core performance. **http://dgit.in/jan21-37**

RTX 3080 Ti & RTX 3060 Ti
NVIDIA RTX 3080 Ti with 20GB VRAM and RTX 3060 Ti with 12GB VRAM leaked. **http://dgit.in/jan21-38**

AMD Radeon RX 6900 XT Graphics Card

A competitor, at last

The AMD Radeon RX 6900 XT is what we have been waiting for so long. A flagship graphics card that towers above all the oompa-loompas. The Indian pricing for the RX 6800 XT was a shocker but its younger sibling, the RX 6800 was optimally priced and retained its position in the GPU hierarchy in line with international markets. The reference card is priced at INR 94,388 (SEP 79,990) which comes as a relief after the disastrous pricing of the 6800 XT.

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FEATURES..................83
PERFORMANCE...........90
VALUE..........................75
DESIGN........................75

Performance

In 3DMark, we prefer the Fire Strike Ultra benchmark since the Extreme and normal runs have started producing ridiculous scores with newer GPUs. The other benchmark within 3DMark which we use is Time Spy and we run both, the normal run and the extreme run. In Fire Strike Ultra the Radeon RX 6900 XT beats the NVIDIA RTX 3090 with a decent margin as it scores well over the 13K mark whereas the RTX 3090 teeters around 12.5K. The Radeon RX 6800 XT and 6800 aren't that far behind. Time Spy also saw the RX 6900 XT maintain a healthy lead over the more expensive RTX 3090 from NVIDIA.

Blender is a great benchmark that makes good use of the GPGPU parallelism within modern GPU microarchitectures. We see the RTX 3090 leading the set followed by the RTX 3080 and RTX 3070. The new RDNA2 cards start off from the 4th position with the RX 6900 XT followed by the RX 6800 XT. There's a decent 20 sec gap between the two competing flagships.

Basemark GPU is a nice benchmark to compare the performance of different graphics APIs between cards. We can use the same textures with OpenGL, Vulkan and DirectX 12 to see if the graphics card excels at any particular API more than the rest or if the performance is consistent across the board. The RX 6900 XT comes third, behind the RTX 3090 and the RTX 3080. NVIDIA cards have a clear advantage in this benchmark and the gap between the RTX 3070 and the RX 6800 has narrowed. What stood out for us was the OpenGL performance as the RDNA 2 cards tanked compared to the competition and are showing only a meagre generational improvement.

In Borderlands 3, we see the 6900 XT equal the RTX 3090 on 1080p at around 125 FPS but as we move into 1440p and 4K resolutions, we see the RTX 3090 take the lead by about 5-7 FPS. Hideo Kojima's Death Stranding see the RX 6900 XT ahead of the RTX 3090 by about 3-6 FPS across 1080p, 1440p and 4K resolutions. In Shadow of the Tomb Raider, we see a similar trend but with frame rates north of 60 FPS since it's a slightly older title. Battlefield V caps out at about 175 FPS in 1080p and 1440p for the RTX 3090, 3080 and the RX 6900 XT and 6800 XT. At 4K, the RTX 3090 gains a little headway with 124 FPS. In Metro Exodus, the 6900 XT is ahead at 137 FPS on 1080p but in 4K we see the RTX 3090 move ahead again. In the remaining titles, we see similar trends. Good thing about the cards is that they're all scoring more than 60 FPS in 4K even in recent AAA-titles.

Verdict

It would appear that AMD has bagged another victory. The AMD Radeon RX 6900 XT is not only great in performance but also has really good pricing. Performance is at par and sometimes even better than the RTX 3090 but the new RDNA2 cards don't have great OpenGL performance. Not that a lot of video games use OpenGL these days, but folks interested in the GPGPU performance might be a bit disappointed with the same. And lastly, there's the ray-tracing benchmark which pegs the RX 6900 XT a little behind the RTX 3090. Visually, they don't seem that far apart so that's fine in our books. For a price of INR 94,388, the AMD Radeon RX 6900 XT is a great 4K gaming card. The downside is that you can not find a card in the market.

–Mithun Mohandas

SPECIFICATIONS
STREAM PROCESSORS: 5120 | **CU** - 80 | **BOOST CLOCK:** 2250 MHz | **VRAM:** 16 GB GDDR6 | **BUS-WIDTH:** 256-bit | **CONNECTORS:** 2x 8-PIN | **TBP:** <300W

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